



GRACE COLLEGE OF ENGINEERING
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Unit- V Probabilistic Reasoning

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Unit- V Probabilistic Reasoning

PART A

1. Define Reinforcement Learning.

This Learning is rather than being told what to do by teacher, a reinforcement learning agent must learn from occasional rewards.

Example If taxi driver does not get a tip at the end of journey, it gives him a indication that his behaviour is undesirable.

2. Define Inductive Learning.

An algorithm for supervised learning is given as input the correct value of the unknown function for particular inputs and it must try to recover the unknown function.

3. Define Classification Learning.

Learning a discrete valued function is called is called classification learning.

4. Define Regression learning.

Learning a continuous valued function is called regression learning.

5. What are the types of machine learning?

i. Supervised Learning ii. Unsupervised Learning iii. Reinforcement Learning

6. Differentiate Supervised and Unsupervised learning.

Supervised Learning It involves learning a function From examples of its inputs And outputs

Example: Applying Brake on the wet road, we can even skip on the road is a result. Unsupervised Learning

It involves learning patterns in the input when no specific output values are supplied.

Example: Day by day agent will learn about “Good traffic days” and “Bad traffic days” without any advice.

7. Define Bayesian Learning.

It calculates the probability of each hypotheses, given the data and makes predictions on that basis, (i.e.) predictions are made by using all the hypotheses, weighted by their probabilities rather than by using just single “best” hypotheses.

8. Define MAP.

Maximum A Posteriori. A very common approximation is to make predictions based on single most probable hypotheses. This is MAP.

9. Define MDL.

The MDL (Maximum Description Length), is a learning method which attempts to minimize the size of the hypotheses and data encodings rather than work with probabilities.

10. What is Maximum – Likelihood hypotheses?

ML – it is reasonable approach when there is no reason to prefer one hypotheses over another a prior. 11. What are the methods for maximum likelihood parameter learning?

- i. Write down an expression for the likelihood of the data as a function of the parameter.
- ii. Write down the derivative of the log likelihood with respect to each parameter.
- iii. Find the parameter values such that the derivatives are zero.

12. Define Naïve Bayes model.

In this model, the “class” variable C is the root and the “attribute” variable X_i are the leaves. This model assumes that the attributes are conditionally independent of each other, given the class.

13. Define sum of squared errors.

The difference between the actual value y_j and the predicted value $(\theta_1 x_j + \theta_2)$ so E is the sum of squared errors.

14. Define EM.

Expectation Maximization: the idea of EM is to pretend that we know the parameters of the model and then to infer the probability that each data point belongs to each component. After that we refit the components to the data, where each component is fitted to the entire data set with each point weighted by the probability.

15. What are the 2 steps in mixture model parameters?

- i. E – Step
- ii. M – Step

16. Define Neural Networks.

It consists of nodes or units connected by directed links. A link propagates the activation. Each link has a numeric weight which determines the strength and sign of the connection.

17. Give Activation function.

$$n_{ai} = g(\text{ini}) = g(\sum W_{j,i} a_j)$$

18. What are the two functions in Neural network's Activation functions?

- i. Threshold function
- ii. Sigmoid function

19. What are the categories of neural network structures?

- i. Acyclic (or) Feed – forward networks ii. Cyclic (or) Recurrent Networks

20. .Define Passive learning.

The agent's policy is fixed and the task is to learn the utilities of states, this could also involve learning a model of the environment.

21. Define Active Learning.

The agent must learn what to do. An agent must experience as much as possible of its environment in order to learn how to behave in it.

22. What are the tasks in probabilistic language model?

- i. Information retrieval ii. Information Extraction iii. Machine Translation

23. What are the types of machine translation?

- i. Rough translation ii. Restricted source translation iii. Pre edited translation iv. Literacy translation

24. How to use the parameters for machine translation?

- i. Segment into sentences ii. Estimate the French language model iii. Align sentences iv. Estimate the initial fertility model v. Estimate the initial word choice model vi. Estimate the initial offset model vii. Improve all the estimates.

25.How are Bayesian networks represented?

Bayesian networks are represented as **directed acyclic graphs (DAGs)**, where:

1. **Nodes** represent random variables (which can be observable or hidden).
2. **Edges** represent conditional dependencies between the variables. An edge from node AAA to node BBB indicates that BBB is conditionally dependent on AAA.
3. Each node has a **conditional probability distribution (CPD)**, specifying the probability of the node given its parents in the graph.

26.What is the purpose of relational probability models?

Relational Probability Models (RPMs) are used to represent and manage uncertainty in complex domains by modeling the probabilistic relationships between objects and their interactions, enabling efficient reasoning, inference, and learning in structured data environments

27. Why does uncertainty arise?

Uncertainty arises due to incomplete information, inherent randomness in events, complexity of systems, measurement errors, and subjective interpretation, making it difficult to predict outcomes accurately.

28. What is Baye's rule? Mention its use.

Bayes' Rule, also known as Bayes' Theorem, is a fundamental principle in probability theory that describes the probability of an event, based on prior knowledge of conditions that might be related to the event. It is expressed mathematically as:

$$P(A|B)=P(B|A) \cdot P(A)P(B)$$

29. what is Bayesian network? Explain the method for constructing

30. Justify the purpose of Bayesian networks.

Bayesian networks are used to model the probabilistic relationships among variables, allowing for efficient reasoning and decision-making under uncertainty. They help in representing complex systems, updating beliefs with new evidence, and inferring outcomes, making them valuable in fields like AI, diagnostics, and risk assessment.

31. What is Causal networks?

Causal Networks, also known as **Causal Bayesian Networks**, are graphical models that represent causal relationships between variables. They are used to visualize and analyze the cause-and-effect relationships in a system. Each node in the network represents a variable, and directed edges between nodes indicate a causal influence from one variable to another.

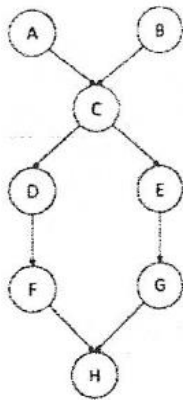
PART-B

1. Derive Baye's theorem with suitable example.
2. Explain the probabilistic reasoning.
3. Discuss the Causal networks with example.
4. Explain about the exact inference in Bayesian network.

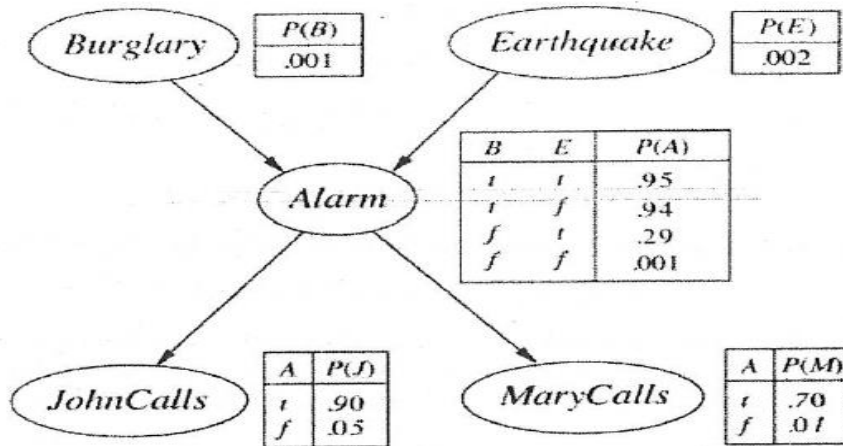
5. Explain the Steps involved in forward and backward algorithm.
6. Discuss the uncertain knowledge with example.
7. Explain about fuzzy set with example.
8. Consider there are 3 boolean variables: toothache, catch, and cavity. From the full joint distribution given below, calculate the following:

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	0.108	0.012	0.072	0.008
$\neg$ <i>cavity</i>	0.016	0.064	0.144	0.576

9. What is d-separation? When are two nodes d-separated? From the Bayesian network given below, find whether D and E are d-separated given evidence about both A and B? State reasons for your answer.



10. You have a new burglar alarm installed at home. It is fairly reliable at detecting a burglary, but also responds on occasion to minor earthquakes. You also have two neighbors, John and Mary, who have promised to call you at work when they hear the alarm. John nearly always calls when he hears the alarm, but sometimes confuses the telephone ringing with the alarm and calls then, too. Mary, on the other hand, likes rather loud music and often misses the alarm altogether. The Bayesian network and the conditional probability table (CPT) for the scenario is given below. In the CPTs, the letters B, E, A, J, and M stand for Burglary, Earthquake, Alarm, JohnCalls, and MaryCalls, respectively. From the Bayesian network, find $P(B|J,M)P(B \mid J, M)$ using variable elimination.



11. (i) What are the ways to understand the semantics of Bayesian Networks?

ii) Discuss the exact inference in Bayesian networks.

12. Design a Bayesian belief network for the diagnosis of car's electrical system.

13. Briefly explain exact inference in Bayesian Networks.